

PRACTICAL - 3

Problem Definition: Design responsive layouts for Home, Registration, Dashboard, Events, and Contact pages using CSS Grid, Flexbox, and Bootstrap. Ensure the website follows a mobile-first responsive design.

Code / Script / Task

Tools and Technology Used

- HTML5 and CSS3
- Bootstrap 5.2.3
- Flexbox and Bootstrap Grid
- Browser responsive mode

Bootstrap layout example

```
<div class="row g-4">
  <div class="col-12 col-md-4">
    <section class="card h-100">...</section>
  </div>
</div>
```

The project uses Bootstrap containers, rows, columns, cards, forms, tables, buttons, and a responsive navbar. A small shared CSS file keeps the black theme consistent without unnecessary inline CSS.

Output: Home Page - Desktop View

 StudentHub desktop home page

Desktop view (1440 px)

The two announcement cards are placed side by side on desktop screens using Bootstrap columns. The navigation links remain in one row.

Output: Tablet and Mobile Views

 StudentHub tablet home page

Tablet view (768 px)

 StudentHub mobile home page

Mobile view (375 px)

On small screens, Bootstrap stacks the content vertically. The navbar collapses into a toggle button to keep navigation readable.

Question & Answer

1. How does the layout adapt for mobile, tablet, and desktop screens?

The website follows a mobile-first approach. On mobile, cards and form fields use the full width. On tablet and desktop, Bootstrap column classes such as **col-md-6** and **col-md-4** display cards in two or three columns. The navigation menu collapses on small screens.

2. Which layout technique is used and why?

Bootstrap Grid is used for page columns because it provides predefined responsive breakpoints. Flexbox is used by Bootstrap for alignment in the navbar and card content. These techniques reduce code and make the layout easier to maintain.

3. Are typography, spacing, colors, and contrast readable?

Yes. The project uses Bootstrap typography and spacing classes for a clean layout. The black navbar, black buttons, white cards, and light background provide clear contrast. Labels and headings use readable sizes.

Question & Answer

4. Is the framework used consistently without unnecessary inline CSS?

Yes. All pages use the same local Bootstrap stylesheet and shared **style.css** file. Bootstrap classes are used for buttons, forms, cards, tables, spacing, and responsive columns. Inline CSS has been removed.

Responsive Pages Implemented

- Home page with announcements and quick resources
- Registration and Login pages with responsive forms
- Student and Faculty Dashboard pages with cards and tables
- Events page with responsive event cards
- Contact, Profile, Courses, Assignments, Attendance, and Admin pages

Basic CSS used in the project

```
body { background: #f2f2f2; color: #222; }  
.hero, .btn-primary { background: #000; }  
.navbar-brand img { width: 32px; height: 32px; }
```

Conclusion / Learning Outcome

The StudentHub website was redesigned as a responsive interface using Bootstrap, Flexbox, and Grid. The pages adapt correctly on mobile, tablet, and desktop screens. Bootstrap components help maintain consistent forms, cards, tables, buttons, and navigation across the project.

This practical demonstrates mobile-first responsive design, readable typography, consistent black color styling, and organized CSS. The final website is simple, reusable, and suitable for a multi-device student portal.

GitHub Link: <https://github.com/d26it118-png/Student-Hub.git>